ENGINEERING MATERIALS

SIZING

Particle Size

These SRMs are used for particle size measuring instruments, including light scattering, electrical zone flow-through counters, optical and scanning electron microscopes, sedimentation systems, and wire cloth sieving devices.

SRM	Particle Diameter Distribution (µm/nm)	Unit Size (g)
Glass Beads, Soda Lime		
1021	2 µm to 12 µm	4
1003c	20 µm to 50 µm (No. 635 to No. 325)	28
1004b	53 µm to 125 µm (No. 270 to No. 120)	43
1017b	106 µm to 355 µm (No. 140 to No. 45)	70
1018b	250 µm to 710 µm (No. 60 to No. 25)	87
1019b	850 µm to 2000 µm (No. 20 to No. 10)	200
Sand		
RM 8010	(No. 30 to No. 325)	3 × 150 g
Silicon Nitride (equiaxed)		
659	0.2 µm to 10 µm	5 × 2.5 g
Zirconium Oxide (irregular)		
1978	0.2 µm to 10 µm	5
1982	10 µm to 150 µm	10
Tungsten Carbide/Cobalt (spheroidal)		
1984	9 µm to 30 µm	14
1985	18 µm to 55 µm	14
Polystyrene Spheres Unit Size: 5 mL vial (unless otherwise noted)		
1690 (0.5 % in H_2O)	0.895 µm	
1691 (0.5 % in H_2O)	0.269 µm	
1961* (0.5 % in H_2O)	29.64 µm	
1963a** (0.5 % in H_2O)	0.1007 µm	
1964 (0.06 µm)	0.0639 µm	5 mL vial
1965 (Slide Mounted: 1 slide)	9.94 µm (hexagonal array) 9.89 µm (unordered clusters)	
Gold Nanoparticles		
8011	10 nm	2 ampoules: 5 mL each
8012	30 nm	2 ampoules: 5 mL each
8013	60 nm	2 ampoules: 5 mL each

*Developed in cooperation with NASA
**This SRM is limited to the calibration of electron microscope and surface scanning inspection systems (not suitable for applications where monosize, unagglomerated spheres are necessary).

Cement Turbidity and Fineness

SRM 46h is only to determine sieve residue according to ASTM C430. Each set consists of 10 sealed vials, each containing approximately 5g of cement.

SRM 114q is for calibrating the Blaine fineness meter according to the latest issue of ASTM C204, to calibrate the Wagner turbidimeter according to ASTM C115, to determine sieve residue according to ASTM C430, and to verify procedure for particle size distribution by a laser diffraction method (no-standard method available). Each set consists of 20 sealed vials, each containing approximately 5g of cement

SRM	Description	Properties Certified	Value	Unit Size
114q	Portland Cement	Residue on 45 µm (No. 325) sieve	0.79 %	set (20)
		Specific Surface Area (Wagner Turbidimeter)	2183 $cm^2\ g^{-1}$	
		Specific Surface Area (Blaine Air Permeability Apparatus) Particle Size Distribution	3818 $cm^2\ g^{-1}$ 1-128 µm	
46h	Portland Cement Fineness Standard	Sieve Residue (45 µm residue) (No. 325)	7.43 %	10 × 5 g

Specific Surface Area (SSA) of Powders (Brunauer, Emmett, and Teller Method)

SRM	Description	Surface Area (m^2/g) Multi-point	Surface Area (m^2/g) Calculated	Single Point	Unit Size (g)
1897	Specific Surface Area Standard	258.32		253.08	7
1899	Specific Surface Area Standard	10.52		10.67	4
1900	Specific Surface Area Standard	2.85		2.79	4
2696	Silica Fume		(22.92)*		70

**The surface area for 2696 was calculated from a combination of single-point, and multi-point calibrations.*

Mercury Porosimetry Standards

SRM	Description	Unit Size (g)
1917	Mercury Porosimetry Standard (Alumina Beads)	10
1918	Mercury Porosimetry Standard (Extruded Silica-Alumina)	12

Particle Count Materials

These SRMs are suitable for use with particle sizing instrumentation, including optical counters, in accordance with National Fluid Power Association (NFPA) T2.9.6 R2-1998 and ISO/DIS 11171.

SRM	Description	Particle Concentration	Unit Size
2806a	Medium Test Dust in Hydraulic Fluid	2.8 mg/L	400 mL
RM 8631a	Medium Test Dust	1 µm to 50 µm	20 g
RM 8632	Ultrafine Test Dust	1 µm to 20 µm	20 g

SURFACE FINISH

Abrasive Wear

This SRM is suitable for use with ASTM G 65, Procedure A.

SRM	Description	Unit Size
1857	D-2 Tool Steel	2 blocks: 0.78 cm × 2.5 cm × 7.6 cm

Surface Roughness

Unit Size: 25 mm × 34 mm × 12 mm

These SRMs are used for calibrating stylus instruments that measure surface roughness. These electroless-nickel coated steel blocks have a sinusoidal roughness profile machined on the top surface.

SRM	Description	Roughness, R_a (µm)	Wavelength, D (µm)
2071b	Sinusoidal Roughness	0.3	100
2073a	Sinusoidal Roughness	3	100
2074	Sinusoidal Roughness	1	40
2075	Sinusoidal Roughness	1	800

FIRE RESEARCH

Surface Flammability

This SRM is suitable for checking the operation of radiant panel test equipment in accordance with ASTM E 162-78.

SRM	Description	Certification	Unit Size (cm)
1002d	Hardboard Sheet	Flame Spread Index, I = 203 Heat Evolution Factor, Q = 42.0	4 sheets: 15.2 × 45.7 × 0.6

Cigarette Ignition Strength Standard

This SRM is intended for use by test laboratories to access and control their testing of cigarette ignition strength in accordance with ASTM Standard Mehods E 2187-04 (or ASTM E2187-02b). The SRM unit consists of one carton of cigarettes containing 10 packs of 20 cigarettes each.

SRM	Description	Ignition Strength	Unit Size (cm)
1082	Cigarette Ignition Strength Standard	12.6%	200 cigarettes - one carton

Smoke Density Chamber

These SRMs are suitable for use with National Fire Protection Agency (NFPA) 258-1998. SRM 1006d is also suitable for use with ASTM E 662-95.

SRM	Description	Maximum Specific Optical Density (D_m (corr.))	Unit Size (cm)
1006d	Non-Flaming Exposure Condition (paper)	193	9 sheets: 17.2 × 25.4 × 0.165
1007b	Flaming Exposure Condition (plastic)	388 to 512	1 sheet: 25.4 × 25.4 × 0.076

Smoke Toxicity

SRM	Description	Combustion on Mode	Observation Time	Values		Unit Size
				LC_{50}	N-Gas	
1048	Cup Furnace Smoke Toxicity Method Standard (ABS copolymer)	Flaming	WE*	27	1.4	8 sheets: (16 × 16 × 0.76) mm
			WE & PE**	25	1.5	
		NonFlaming	WE*	58	1.2	
			WE & PE**	53	1.4	
1049	University of Pittsburgh I Smoke Toxicity Method Standard (Nylon 6/6)		30 min exposure, plus 10 min post-exposure	4.4		150 g

**WE = within 30 minutes*
***WE & PE = 30 minutes + 14 days*

Flooring Radiant Panel

This SRM is suitable for use with ASTM E 648-78 and NFPA 253-1978.

SRM	Description	Critical Radiant Flux	Unit Size (cm)
1012	Flooring Radiant Panel (Kraft Paperboard)	0.36 W/cm^2	3 sheets: 104.1 × 25.4 × 0.305

NONDESTRUCTIVE EVALUATION

Artificial Flaw for Eddy Current NDE

RM	Description	Flaw Size	Unit Size
8458	Artificial Flaw (Aluminum Alloy)	3.0 mm × 0.1 mm	7 cm × 7 cm × 2 cm

PERFORMANCE ENGINEERING MATERIALS

Fracture Toughness of Steels (Charpy V-Notch Test Blocks) and Izod Impact

Unit Size: set of 10 mm × 10 mm × 54 mm specimens

These SRMs are suitable for use with ASTM E 23 and ISO/DIS 12736.

SRM	Description	Energy Range (J)
2092	Low Energy (4340 Alloy Steel)	13 to 20
2096	High Energy (4340 Alloy Steel)	88 to 136
2098	Super High Energy (Maraging Steel)	176 to 244
2115	Low Energy Izod	13 to 25

Rockwell Hardness

Unit size: 60 mm diameter × 15 mm

SRM	Description	Nominal Hardness (HRC)
2810	Rockwell C Scale Hardness - Low Range	25
2811	Rockwell C Scale Hardness - Mid Range	45
2812	Rockwell C Scale Hardness - High Range	63

Microindentation Hardness (Knoop and Vickers Test Blocks)

Unit Size: 1.15 cm × 1.15 cm (unless otherwise noted)

These SRMs are suitable for use with ASTM E 384.

SRM	Description	Load (N)	Hardness (kg/mm^2)
Copper, Bright			
1893	Knoop	0.245, 0.49, 0.98	125
Nickel, Bright			
1894a	Vickers	0.245, 0.49, 0.98	125
1895	Knoop	0.245, 0.49, 0.98	600
1896b	Vickers	0.245, 0.49, 0.98	600
1905	Knoop	2.943	600
1906	Knoop	4.905	600
1907	Knoop	9.81	600
1908	Vickers	2.943	500
1909	Vickers	9.81	500
2798a	Vickers	4.905	600
Silicon Nitride, Ceramic			
2830 (22 mm diameter × 9.54 mm)	Knoop	19.6	1500
Tungsten Carbide, Ceramic			
2831 (25 mm diameter × 9.5 mm)	Vickers	9.8	1530

Tape Adhesion Testing

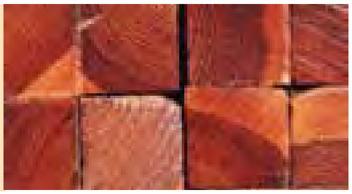
This SRM is suitable for use with ASTM D 2860 and ASTM D 3654.

SRM	Description	Unit Size
1810a	Linerboard for Tape Adhesion Testing	50 sheets: 21.6 cm × 28 cm

Bleached Kraft Pulps

These RMs are intended primarily for use in fundamental studies on the physical properties of fibers and paper sheets. No extensive property measurements have been made on these materials beyond ensuring that they were within the control limits of the normal production run.

RM	Description	Unit Size
8495*	Northern Softwood	10 standard lap sheets: 0.5 kg each
8496*	Eucalyptus Hardwood	10 standard lap sheets: 0.5 kg each



**Developed in cooperation with the Pulp Material Research Committee*

Secondary Ferrite Number (FN) Materials

The RMs are suitable for use with ANSI/AWS A4.2 and ISO 8249.

RM	Ferrite Number	Unit Size (mm)
8480	0 to 30	10 × 12 × 20
8481	30 to 120	10 × 12 × 20

Fracture Toughness of Ceramics

Unit Size: 3 mm × 4 mm × (45 to 47) mm

SRM	Description	Fracture Toughness (MPa · $m^{1/2}$)	No. of Specimens
2100	Silicon Nitride Flexure Specimens	4.57	5

Magnetic Moment Standards

SRM	Description	Certified Property	Unit Size
762	Nickel Disk	Specific Magnetization	disk: 6 mm diameter × 0.13 mm
764a	Magnetic Susceptibility Standard Platinum	Magnetic Moment	cylinder: 2 mm diameter × 3.42 mmL
772a	Nickel Sphere	Magnetic Moment	sphere: 2.383 mm diameter sphere
2853	Yttrium Garnet Sphere	Magnetic Moment	sphere: 1 mm diameter (2.8 mg)